UNIVERSITÄT ZU KÖLN

Supply Chain Management Area

Supply Chain Analytics I: Predictive Analytics

Project: Brooklyn Group for Consulting

Day 1 It's Monday morning, and you're in a hurry. You can't afford to be late for your first meeting—it's your first day as a trainee at the New York office of the Brooklyn Group for Consulting (BGC). What started as a small firm by the Brooklyn Bridge has evolved into a global consulting powerhouse with over 100 offices worldwide. You're keen to make a strong first impression.

As a trainee at BGC, you'll rotate through various projects before settling on one by the end of the program. You check your inbox, and the first assignment is on the project *Business Analytics at Citi Bike*, New York City's bike-sharing service. With your background in Supply Chain Management and your university analytics course, you feel well-prepared to impress.

Still, there's a hint of nerves as you approach the conference room. Just as you're about to enter, someone waves you over. "So, you're the new trainee, huh?" says the man in bermuda shorts, flip-flops, and a Hawaiian shirt. This is Clint, better known as "The Guru" around the office. No matter the season, The Guru dresses like he's on his way to Miami Beach. When you asked your colleagues about his nickname, they told you that when it comes to business analytics, the Guru is unmatched—and he's going to be your mentor.

The Guru points you to your seat, but before he can say anything, the managing director (MD) walks in.

The MD gets straight to the point. She greets everyone, welcomes you to the team, and dives into the task at hand.

"Guru, we've finally received a dataset from Citi Bike. I'll email you the details later, but for now, just work your magic. Maybe our new trainee can help you out?"

The Guru nods, smiling. His enthusiasm is palpable—it's as if he's just received an early Christmas gift. You finish your coffee in one gulp, bracing yourself for what promises to be a long day.

Later, you're in The Guru's office, watching him read the MD's email with the dataset and some preliminary questions.

"You know what I've learned during my time at BGC?" he asks, but doesn't wait for your reply. "Before diving into any complex analysis, you need to get a feel for the dataset." He pauses thoughtfully. "I think this is a perfect task for you. Are you ready for it?"

You nod, eager to get started. The Guru advises you to settle in at your desk, promising to add some helpful tips to the MD's email before forwarding it to you. He reassures you that his door is always open if you have questions.

As soon as you're set up, you receive the MD's email, now annotated with The Guru's notes in italics.

Subject: Business Analytics at Citi Bike

Hi Guru,

Attached, as promised, my specific questions regarding the data set.

Aside from those, please look into the data, and see what you can find.

a) *Create an overview table that contains a short description of each variable (values the variable can take, number of missing values, etc.)*
b) *How many days worth of data have we got?*
How many hours are there in a day? For example, if we have 48 observations, then ...
c) *Do we have the same number of observations for each hour and day?* This is a smart question. What would the implication be for, say, the observed average number of rides per hour if the hours in which no rides took place were simply missing from our data set?
d) *Which variables have the strongest correlation with the number of rides?*
e) *Create four meaningful diagrams (e.g. a bar chart, a histogram, a boxplot, a scatter plot) and briefly explain what they show.* If you don't know where to start: I have the impression that (i) people rarely cycle in winter months, and that (ii) Mondays will have the most number of rides, because people are often running late after the weekend.
f) *Create a diagram that shows*

    1. *The average temperature per day over the time horizon*
    2. *The total number of rides per day over the time horizon*

    in the same frame and interpret the result.

Day 2 On your second day at the office, you find The Guru already at his desk. His outfit is in different colors today, but the signature style remains unchanged. Rumor has it that he owns a different Hawaiian shirt for every day of the year—a claim that's starting to seem entirely believable.

The Guru is reviewing the email you sent him yesterday with your findings. He's impressed with your work and plans to forward your results to the MD.

Today, The Guru wants to build a model to identify the factors influencing the number of hourly rides.

To get started, he asks you to...

a) ... fit two reasonable regression models and to interpret your results. (For example, does your model predict more trips per hour on weekends? More trips if it is sunny? . . .) No independent variable should appear in both models.
b) ... predict the number of rides for an average hour in July at noon with both models. Do they differ in their prediction?
Note that, depending on your choice of variables, you may first have to calculate the mean values of the respective independent variables for the respective hour.
c) ... plot the residuals, and to describe what you see.

Before you start working on this task, he gives you the following rules and tips:

i. **Do not** split the data into a test-train set.
ii. You may use whichever variables you see fit.
iii. Remember that some variables are categorical. To reduce the number of coefficients in your model, you are free to condense them. For example, if you don't want to include a dummy for each weekday, you could create a binary variable that is equal to one if it is the weekend, and zero else.
iv. Take a look at the model predictions you get for the number of hourly rides in December at midnight. Do you see any issues?

Day 3 It's hard to believe it's only Wednesday—it's just your third day at BGC, but you're already feeling like part of the team. You also can't help but notice that The Guru is wearing yet another Hawaiian shirt.

"Ah, good morning," The Guru says, glancing up from his laptop. "Just the person I was looking for. We've got a new project titled *Predictive Maintenance*. That's what I love about consulting: there's always something new, and the project names are always intriguing."

The Guru explains that a manufacturing company wants to enhance its quality assurance by developing a model that uses production data to predict if a part is faulty.

"To this end," he continues, "they've provided us with a dataset containing various production process characteristics, like temperature, a binary variable indicating if the part was made on a Monday, and, of course, information on whether the part is faulty."

"I really liked the work you did on Citi Bike. So please:

a) Come up with a classification model this time. To get a feel for the data, you may use the full data set. But to see how well your model performs, please use the test-train split I've already created.
b) Briefly interpret your final model. Which factors seem to be most important?

"Thanks, Trainee," The Guru says.

As you're about to leave, he mentions that he's a big believer in healthy competition—so much so that he's already built a model for you to benchmark against. On the test set, his model achieves an accuracy of 93.5%, a precision of 35%, and a recall (true positive rate) of 67%..[1]

He challenges you to build a better model. For the purposes of comparison, The Guru states that a model is undeniably better if it simultaneously achieves a higher accuracy, higher precision, *and* a higher recall.

Before you start working on this task, he gives you the following rules and tips:

i. You must employ a classification model, and you must make use of the test-train split provided. Other than that, there are no restrictions.
ii. The Guru recommends to start just like before, namely by taking a closer look at the data.
iii. Note that we have two dependent variables in the data set: *Target*, a binary variable which is equal to one only if the part is faulty, and *Failure Type*, which provides information on the exact type of failure. The Guru ignored *Failure Type*, and only used *Target* as the dependent variable in his model. He recommends you do the same.
iv. If you want to visualise a confusion matrix, The Guru recommends using the code which he has posted in the project folder on ILIAS. Note that this confusion matrix also automatically computes all of the performance metrics you may be interested in, such as accuracy, precision, and recall.

[1] If you need to refresh your memory as to what these classification metrics mean, The Guru suggests taking a look at the Wikipedia page **here**.

Day 4 When you arrive at the office the next morning, you're relieved to see that The Guru took it in stride that you outperformed his model. He admits his defeat with a smile and hands you a coffee to celebrate.

"In other news, the MD emailed me this morning. She had a chat with Citi Bike. . . " The Guru says, taking a sip of his coffee.

"What's the ask?" you inquire cautiously.

The Guru scratches his head. "Nothing too wild. Someone at Citi Bike heard about *LASSO regression* and now they want us to run one on the entire dataset with all the available variables. In the end, they're interested in seeing which coefficients the LASSO regression selects." They would like us to use three different tuning parameters $\lambda = 0$, $\lambda = 2$, and $\lambda = 4$.'

'Ok, should I get that done then?'

'If you feel you're up to it, please.'

Hint: Can you answer the question for $\lambda = 0$ without running a model?

Day 5 As you step into the office on Friday morning, you notice that The Guru is already hard at work. For the first time, you spot a postcard pinned to the wall behind him that reads, "Yesterday, I ate a clock. It was very time-consuming. Especially when I went back for seconds."

You're about to comment, but The Guru speaks first.

"Good morning," he says, smiling through the blinding colors of his Hawaiian shirt. "Looking forward to the weekend?"

"Oh, definitely," you reply. "But I have a feeling you've got another task lined up for me?"

"We have a winner!" The Guru exclaims, mimicking the sound of a ringing bell with his hand. "And I think you'll enjoy this one." He leans back in his chair before explaining, "You know how BGC partners with a few local universities? Well, one of the professors held an auction during a course, and they want us to analyze the data. I've already emailed you the details."

You nod in acknowledgment.

"Once you've finished that, I'd say call it a day."

You nod again and head toward the door.

"And remember: TGIF," The Guru adds, still grinning as you leave his office.

Subject: Auctions

Hi Trainee,

Attached, the auctions data set. You will find a description of the auctions below.

We need to address the following points.

a) *Use the data to derive an empirical distribution of the auction price for both the two-bidder auction and the four-bidder auction. To do so, simulate 1,000 two-bidder and four-bidder auctions. Plot the results.*
b) *With which probability is the price above 8 in a two-bidder auction, and in a four-bidder auction, respectively?*
c) *Instead of first-price auctions, you can also run second-price auctions. In a second-price auction, the (equilibrium) price is equal to the second-highest valuation among participants. Simulate 1,000 two-bidder and four-bidder auctions using participants' valuations. Illustrate the price distributions for the two and four bidder cases with histograms.*
d) *Do first- and second-price auctions with two bidders yield the same expected prices? Does the variance of prices differ? Conduct the same analysis for auctions with four bidders.*[2]

TGIF,

The Guru

[2] As a side-note: this question is motivated by a surprising result in auction theory, the so-called *revenue equivalence theorem*. Loosely stated, the theorem states that any auction yields the same expected revenue, i.e. price, to the auctioneer. Specifically, the expected price from a first-price auction is *identical* to the expected price from a second-price auction.

Description of the first-price auction: Consider a group of students bidding for a good. A student's birthday determined their valuation v for the good in the following way:

$$v = (dd + mm) \cdot 0.3$$

For example, if December 31st is your birthday, your valuation v is equal to $(31 + 12) \cdot 0.3 = 12.9$.

After determining their valuations, the students were asked to write down two numbers:

a) Their bid in a two-bidder auction, and
b) Their bid in a four-bidder auction.

The rules of the auction are as follows: the highest bidder wins, and the price is given by their bid. This is what we call a *first-price auction*.

Description of the second-price auction: The setting is just as above, and the highest bidder still wins. The only difference is that in a second-price auction, as the name suggests, the price is given not by the highest bid, but by the *second-highest* bid.

Note that in a second-price auction, it is weakly dominant for bidders to bid their true valuations.

Day 6 It's a cold Saturday morning. It's been a long week. You're walking through central park, trying to unwind. Out of habit, you refresh your emails. You instantly regret it.

Subject: Business Analytics at Citi Bike

Hi Trainee,

Great work on the auctions project!

Your week started with Business Analytics at Citi Bike, *so it seems only fitting that it should end with that too.*

Citi Bike would like us to do a few more things. Namely:

a) *Fit a regression tree on the entire dataset using all the variables available to us. I leave it to your expertise when it comes to deciding on things like the maximum depth of the trees, but ask you to interpret the parameter(s).*
b) *Also, Citi Bike would like us to apply a random forest model.*
c) *Ultimately, they want us to determine the feature importances of these two models. How do the results compare to the coefficients we obtained from the LASSO regression with* $\lambda = 4$ *on Thursday?*

Please have the results ready for me by Monday morning.

Have a great one!

The Guru

Day 7 You worked through the entire Sunday. Just as you want to send your results to the Guru, you find another short mail from him:

> *Hi, I just went out with a friend from corporate. She said they want to create another unit similar to ours. They'll probably call you for some interview about your experience. Can you maybe write a short report about your personal experience throughout the last days? Specifically, please answer the following questions: In your opinion, what was the largest obstacle when working on this project? What is the strongest impression the project left on you? Is there anything you think that should be improved? Thanks a lot!*

(Hint: Please use the template provided on ILIAS. You do not need to write more than half a page. Each member of your group needs to write this report individually. Please save it in a .pdf file named in the fashion FirstName_LastName_MatriculationNumber.pdf (e.g. Max_Mustermann_-4567890.pdf). You can still upload them along with your Jupyter Notebook in a group submission.)